CIS 517: Data Mining and Warehousing

Group Project

[1] Project Goals

To apply different data mining techniques in Python to preprocess data, classify it, and evaluate the performance of different classification algorithms.

[2] Final Report Submission Guidelines

Total Marks: 15 marks.

Numbers of students per group: 6-7 students.

Submission Deadline: 4th December 2026, end of day.

Submission: via Blackboard.

[3] Deliverables

Each group must submit the following:

- **Final Project Report**
  - A complete report covering all project tasks and requirements and explaining the main code.
  - Relevant tables, figures, and visualizations should be included where appropriate.
- **Python Code**
  - Submit the complete Python code used to perform all project tasks.
  - The submitted code must be the same code used to produce the results presented in the report.
  - The code must contain clear and meaningful comments explaining the main sections, important processing steps, models, parameters, and evaluation procedures.
  - Code should be organized, readable, and executable.
  - Students must be able to explain the submitted code and the decisions made during its development.
- **Plagiarism Report** with similarity below 20%.
- **AI Usage Declaration Form.**

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[4] Project Specification

You are provided the following **dataset**:

Online Shoppers Purchasing Intention Dataset - UCI Machine Learning Repository

Apply the instructions given in the table to the provided dataset to complete the project requirements:

No.	Task	Marks
1	Use different Python packages including pandas, numpy and other models to analyze the given data and evaluate the performance of the different models. Explain your code.	2 marks
2	Evaluate the dataset using different statistical methods and understand the nature of the data in different attributes.	2 marks
3	Apply data preprocessing to prepare the data for analysis by different models. Explain the steps you took to perform this task.	2 marks
4	Apply an outlier analysis solution to identify outliers among the data. Evaluate these outliers and explain the steps you have taken to address their presence.	2 marks
5	Apply logistic regression to the dataset and the different performance enhancement techniques to fine tune your model. Apply decision trees and random forests to the data to classify the data in the optimal manner and tune the performance.	3 marks
6	Compare the performance with other models you developed as applicable using techniques you learnt in the lab.	2 marks
7	Student team should provide a task table at start of the document highlighting clearly the contribution of each team member to the allocated tasks.	2 marks
	Total Marks	**15**

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[5] Plagiarism and academic dishonesty

- All work submitted by the student teams must be their own original contribution.
- AI tools may be used only for limited support and must not replace students' original work.
  - **Allowed uses:** grammar/language polishing, brainstorming, explanations of course concepts, debugging your own code, and verifying information.
  - **Not allowed:** generating full report sections, producing fake/autogenerated work, or submitting code you cannot explain.
  - **AI Declaration:** If AI is used, students must declare the tool and purpose, complete the AI Usage Declaration Form, and include all prompts in an appendix with the related task or report section.
- Students should submit their plagiarism report along with the original report, with plagiarism count being less than 20%.

[6] Project Rubrics

Task 1		
2 marks	**1.5 – 1 mark**	**< 1 mark**
Students demonstrate correct use of different Python packages including pandas, numpy and other models to analyze the given data and evaluate the performance of the different models	Students demonstrate the use of some packages but do not apply the correct ones in the right places	Students do not use python packages as required.

Task 2		
2 marks	**1.5 – 1 mark**	**< 1 mark**
Students evaluate the dataset using different statistical methods and understand the nature of the data in different attributes.	Students evaluate the dataset using different statistical methods, but they do not apply the correct methods to the attributes	Students perform basic or no evaluation of dataset using statistical methods. Their inferences are not correct

Task 3		
2 mark	**1.5 – 1 mark**	**< 1 mark**
Students apply different preprocessing techniques to the dataset and explain what they did clearly.	Students apply some data preprocessing techniques to the dataset but some discrepancies are not addressed, or explanation is not complete.	Students apply basic data preprocessing techniques only and do not explain clearly

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Task 4		
2 marks	**1.5 – 1 mark**	**< 1 mark**
Students apply outlier analysis to correctly identify the outliers among the data and explain the steps they took to address their presence.	Students correctly identify some outliers but do not explain the steps taken to address them	Outlier identification is vague or incorrect and steps are not taken to address them.

Task 5		
3 marks	**2.5 – 1 mark**	**< 1 mark**
Students apply logistic regression to the dataset and the different performance enhancement techniques to fine tune their model. They also apply decision trees and random forests to the data to classify the data in the optimal manner and tune the performance.	Students apply logistic regression, decision trees and random forests but they do not fine tune the performance for all models.	Students apply some of the mentioned models but not all of them and do not show or explain how they have fined tuned the models

Task 6		
2 marks	**1.5 – 1 mark**	**< 1 mark**
Students compare the performance with other models they developed as applicable.	Students provide some comparison of performance but it may not be meaningful in all places	Students provide a vague comparison which is incomplete.

Task 7		
2 marks	**1.5 – 1 mark**	**< 1 mark**
Student team provides a task table at start of the document highlighting clearly the contribution of each team member to the allocated tasks.	Student team provides the task table, but it does not show the detailed distribution of tasks.	Students show a vague distribution of tasks in the table.